MINOR PROJECT / BOOSTING

Obesity Level Classification

A multiclass machine-learning model using tuned XGBoost

DATASET Obesity Levels - Kaggle https://www.kaggle.com/datasets/fatemehmehrparvar/obesity-levels

PROBLEM & OBJECTIVE

Predict one of seven obesity-level categories from demographic, eating-habit, family-history, and physical-condition features. This is a supervised classification exercise, not a medical diagnosis tool.

DATASET & DATA QUALITY

2,111 rows and 17 columns at load. The notebook found and removed 24 duplicate rows, leaving 2,087 records; no missing values were reported. The target, **NObeyesdad**, has seven classes.

EDA HIGHLIGHTS

TARGET CLASSES

After duplicate removal, class counts are fairly close: Obesity_Type_I is largest (351; 16.8%); Insufficient_Weight is smallest (267; 12.8%).

GENDER & AGE

The cleaned data has 1,052 male and 1,035 female records. Ages span 14-61; mean age is about 24.35 and median about 22.85.

RELATIONSHIPS EXPLORED

Visualizations compare target class with gender, family history of overweight, and frequent high-calorie food consumption (FAVC).

WORKFLOW & MODEL SELECTION

01 PREPARE
One-hot encode predictors; encode the seven-class target.

02 SPLIT & BALANCE
80/20 stratified split; SMOTE only on training data.

03 TUNE XGBOOST
12 candidates; stratified 3-fold CV; maximize macro F1.

04 TEST ONCE
Evaluate on the untouched 418-record test set.

Best CV macro F1: 0.9446 | XGBoost settings: 250 trees, depth 6, learning rate 0.03, subsample 0.9, min child weight 3, gamma 0.0.

HOLDOUT TEST PERFORMANCE

95.45%	95.31%	95.45%	0.9984
ACCURACY	MACRO F1	WEIGHTED F1	ROC-AUC
399 of 418 correct	Unweighted class average	Support-weighted average	Weighted, one-vs-rest

Macro precision: 0.9528 Macro recall: 0.9536 Weighted precision: 0.9548 Weighted recall: 0.9545

CLASS-LEVEL RESULTS

Target class	Precision	Recall	F1-score	Support
Insufficient_Weight	**0.98**	**1.00**	**0.99**	**53**
Normal_Weight	**0.90**	**0.93**	**0.91**	**57**
Obesity_Type_I	**0.99**	**0.96**	**0.97**	**70**
Obesity_Type_II	**0.98**	**1.00**	**0.99**	**60**
Obesity_Type_III	**1.00**	**0.98**	**0.99**	**65**
Overweight_Level_I	**0.91**	**0.87**	**0.89**	**55**
Overweight_Level_II	**0.92**	**0.93**	**0.92**	**58**

Interpretation: Recall is strongest for Insufficient_Weight and Obesity_Type_II (1.00 each). Overweight_Level_I is the hardest class to recover (recall 0.87), indicating confusion with neighboring weight categories.

CONCLUSION & LIMITATIONS

The tuned XGBoost model achieved 95.45% test accuracy with a weighted ROC-AUC of 0.9984, showing strong separation across the seven dataset labels. The confusion matrix and per-class scores still reveal errors, especially for Overweight_Level_I. Results depend on this dataset and its sampling; they do not establish clinical validity or guarantee performance on new populations. Future work should test external data and monitor class-wise errors.